United States Patent	12412024
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Kandari; Pragya et al.

Enhanced text wrapping in documents

Abstract

Embodiments described herein facilitate text wrapping around a text-based glyph in an effective and efficient manner. In embodiments, a text-based glyph is generated within a document. The glyph is generated from a font file that includes glyph data that instruct a document editor how to generate the glyph. Using the glyph data, an intersection between the glyph and a text tile, e.g., a text area boundary where text can be placed, of the document. An insertion location based on intersection is identified, and text is inserted within the text tile at the insertion location, thus providing the effect of wrapping the text around the text-based glyph.

Inventors: **Kandari; Pragya (Noida, IN), Singh; Anurag (Noida, IN)**

Applicant: **ADOBE INC.** (San Jose, CA)

Family ID: **94372263**

Assignee: **Adobe Inc. (San Jose, CA)**

Appl. No.: **18/358743**

Filed: **July 25, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250036854 A1	Jan. 30, 2025

Publication Classification

Int. Cl.: **G06F40/109** (20200101); **G06F40/189** (20200101)

U.S. Cl.:

CPC **G06F40/109** (20200101); **G06F40/189** (20200101);

Field of Classification Search

CPC: G06F (40/109); G06F (40/189); G06F (40/103)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6081816	12/1999	Agrawal	715/210	G06F 40/103
7315979	12/2007	Walker	358/1.15	G06F 3/0483
11170156	12/2020	Arora	N/A	G06F 3/04847
2010/0174985	12/2009	Levy	715/244	G06F 40/163
2011/0179345	12/2010	Capela	715/209	G06F 40/103

OTHER PUBLICATIONS

Bateman et al., “Towards Constructive Text, Diagram, and Layout Generation for Information Presentation” Computational Linguistics, vol. 27, No. 23, copyright 2001 ACM, pp. 409-449. (Year: 2001). cited by examiner

Chao, “Text Block Geometric Shape Analysis” DocEng'06, Oct. 10-13, 2006, Amsterdam, The Netherlands, Copyright 2006 ACM, pp. 22-24. (Year: 2006). cited by examiner

Harkins, S., “How to use the many text wrapping options in Microsoft Word”, TechRepublic.com, Published on Apr. 12, 2021, Retrieved from Internet URL: https://www.techrepublic.com/article/how-to-use-the-many-text-wrapping-options-in-microsoft-word/, accessed on Aug. 14, 2023, pp. 11. cited by applicant

“Wrap text around objects”, Adobe, last updated on Jun. 20, 2023, Retrieved from Internet URL: https://helpx.adobe.com/indesign/using/text-wrap.html, accessed on Aug. 14, 2023, pp. 14. cited by applicant

“Textwrap—Text wrapping and filling”, Python, Retrieved from Internet URL: https://docs.python.org/3/library/textwrap.html, accessed on Aug. 14, 2023, pp. 5. cited by applicant

“Glyf : Glyph Data”, Glyph Data Table, Retrieved from Internet URL: https://fonttools.readthedocs.io/en/latest/ttLib/tables/_g_L_y_f.html, accessed on Aug. 9, 2023, pp. 5. cited by applicant

“Glyf—Glyph Data Table”, Glyf data table (OpenType 1.9)—Typography | Microsoft Learn, Published on Dec. 9, 2021, Retrieved from Internet URL: https://learn.microsoft.com/en-us/typography/opentype/spec/glyf, accessed on Aug. 9, 2023, pp. 12. cited by applicant

“Calculating intersection point of quadratic bezier curve”, Stack Overflow, Retrieved from Internet URL: https://stackoverflow.com/questions/27664298/calculating-intersection-point-of-quadratic-bezier-curve, accessed on Aug. 9, 2023, pp. 6. cited by applicant

“The OpenType Font File”, (OpenType 1.9)—Typography | Microsoft Learn, Published on Dec. 9, 2021, Retrieved from Internet URL: https://learn.microsoft.com/en-us/typography/opentype/spec/otff, accessed on Aug. 9, 2023, pp. 16. cited by applicant

---

Primary Examiner: Nguyen; Phuong H

Attorney, Agent or Firm: Shook, Hardy & Bacon L.L.P.

Background/Summary

BACKGROUND

(1) Text wrap is a functionality provided by a word processing software that allows users to surround a graphic with text. Text wrap causes text to wrap around the graphic so that the graphic does not interfere with line spacing.

SUMMARY

(2) At a high level, aspects of the technology wrap text around a text-based glyph, which allows tight text wrapping along glyph edges that are calculated from glyph data. This is in contrast to wrapping text around an image-based object, where image boundaries often do not represent the edges of an object within the image, and which are typically square or rectangular in nature. In practice, this allows a user to easily change the text-based glyph, and the text wrapped around it, without having to edit image boundaries for an image-based object along which the text is wrapped.

(3) In one example operation, a glyph is generated within a word document using a document editor. The glyph may be any text-based glyph generated from a font file, such as a language character, emoji, and so forth.

(4) The font file provides glyph data from which the document editor can generate the glyph. The instructions provided by the glyph data comprise information, such as control points, curves, and other instructions that define the glyph, including the outline or edge of the glyph.

(5) Text tiles, e.g., text area boundaries, are located within the document and identify locations within the document where text can be placed. Using the glyph data, the locations where the text tiles intersect the glyph are determined. An insertion location, where text is inserted within the text tile, is located based on the intersection. Text is inserted at the insertion location, and the inserted text extends within the text tile away from the glyph. This gives the text the appearance of wrapping around the glyph.

(6) This summary is intended to introduce a selection of concepts in a simplified form that is further described in the Detailed Description section of this disclosure. The Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be an aid in determining the scope of the claimed subject matter. Additional objects, advantages, and novel features of the technology will be set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the disclosure or learned through practice of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology is described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. **1** illustrates an example operating environment in which aspects of the technology can be employed, in accordance with an aspect described herein;

(3) FIG. **2**A illustrates a document having an example image-based glyph wrapped by text, in accordance with an aspect described herein;

(4) FIG. **2**B illustrates a document having an example text-based glyph wrapped by text, in accordance with an aspect described herein;

(5) FIGS. **3**A-**3**C illustrate an example document showing text within respective text tiles, in accordance with an aspect described herein;

(6) FIG. **4**A illustrates an example glyph and a corresponding text tile intersecting the glyph, in

accordance with an aspect described herein;
(7) FIG. **4**B illustrates the example glyph of FIG. **4**A having text inserted within the text tile, in accordance with an aspect described herein;
(8) FIG. **5**A illustrates an example glyph and a corresponding plurality of horizontally offset text tiles intersecting the glyph, in accordance with an aspect described herein;
(9) FIG. **5**B illustrates the example glyph of FIG. **5**A having text inserted within the plurality of text tiles, in accordance with an aspect described herein;
(10) FIGS. **6**A-**6**B illustrate an example in which text tiles are positioned within various regions of a glyph, in accordance with an aspect described herein;
(11) FIGS. **7-13** illustrate various documents having various examples of glyphs wrapped by text, in accordance with aspects described herein;
(12) FIG. **14** illustrates a block diagram having an example method of wrapping a text-based glyph with text, in accordance with an aspect described herein;
(13) FIG. **15** illustrates a block diagram having another example method of wrapping a text-based glyph with text, in accordance with an aspect described herein; and
(14) FIG. **16** illustrates an example computing device suitable for implementing aspects of the technology, in accordance with an aspect described herein.

DETAILED DESCRIPTION

(15) Text wrap is the functionality provided by word processing software that allows text to surround a graphic or object within a document. Conventional methods of text wrap generally include text surrounding a target object that is in a graphic format. These conventional methods are manual and tedious, and in many cases, require incremental manipulation of images or text. These manipulations are often visually based changes, which can lead to errors, particularly at high resolutions or increased scale.
(16) To wrap text around a glyph using conventional methods, the glyph is converted to an image, and the text is wrapped around the boundaries of the image. In most cases, the image boundaries (e.g., rectangular) on which the text is wrapped is not the same as the edges of the glyph (e.g., non-rectangular). As a result, text is not wrapped tightly with the glyph edges, leading to text that is generally wrapped in a rectangular manner.
(17) An existing workaround to this problem is to create an outline for a boundary around an object, such as the glyph within the image, so that the image boundaries are roughly the same as the object's boundaries. The outline is represented as a vector. In such a case, the letter could be treated as a graphic using the outline defined around the glyph. This technique is typically applied after the text is in a desired form. Otherwise, changes to the text generally requires a change to the glyph, thereby resulting in a restructuring of the boundary and representing the new glyph as a new image graphic. Similarly, when modifying a font, the font modification can cause the letter to be represented as a different shape, thus needing different image boundaries. Further, a change to the font size would typically require rescaling of the image-based glyph, which can also result in modification of the previously determined boundaries for the image based on the glyph's edges.
(18) Moreover, when objects, such as glyphs are represented as an image for wrapping text, it often limits or eliminates the ability to apply or change text-based properties to the object in the image. For instance, a user might not be able to apply a new font or font size, or other property, such as bold, italic, or underline, to the object. Instead, the object must be redrawn or modified using an image editor, have new image boundaries defined to match the changes, and reintroduced back to a document as an image-based glyph.
(19) The problem is further aggravated when the fonts used in the document are a decorative type script. In these cases, there may be no lookalike graphic to use as an image within the document editor. As such, defining the image-based edges of the glyph with these decorative scripts is a time-consuming, manual method. Further, glyph modifications, such as changing the sizing or font, generally requires reapplication of the manual process to define new image boundaries based on the

glyph edges.

(20) In many cases, such as publishing books, changing the text when it is represented as a graphic also changes the amount of space available for text. This could lead to other spacing issues, as text is reoriented within the document.

(21) Typically, with conventional word processing software systems, when making changes to image-based glyphs, an external application for designing object properties is usually needed. The object designed or modified in these external image-editing applications is imported to the word processing software. This too has drawbacks, as any changes to the object, such as an image-based glyph, desired in the word processing software must be made in the external application and then reimported into the word document edited using the word processing software.

(22) In cases where an external application is used to make changes, it is difficult to prevent the text from going inside the vector bounds, such as those defining interior features of a glyph. Further, this method typically does not provide the precise control to include or exclude inner edges of an image-based glyph. As such, additional manual changes within the word processing software are typically required to make the necessary adjustments after importing the image into the document.

(23) Moreover, aspects of the present technology reduce the document size relative to conventional methods. In conventional image-based methods, an image-based glyph of a quality suitable for document publication requires significantly more storage space than a text-based glyph, since the text-based glyph is calculated from glyph data as opposed to rendered as an image. This problem is exacerbated when increasing the size or quality of a glyph, since doing so to an image-based glyph significantly increases file size, while doing so to a text-based glyph generally does not. Compared to conventional methods, this preserves computational resources when processing such documents, and further preserves storage space on memory devices.

(24) Accordingly, the technology described relate to live text wrapping to textual features in an automated manner. In this way, text wrapping is applied to text itself (e.g., a text-based glyph) in an efficient and effective manner. In particular, to apply live text wrapping, text is wrapped around a text-based object determined by glyph data from a font file. The text wrapping features described herein allow users to continuously edit a document, yet change or edit textual features around which text wrapped. Thus, instead of wrapping around the boundaries of an image (e.g., in a rectangular manner), the edges of a text-based glyph can be calculated and used to determine where to place the text surrounding the glyph (e.g., in a non-rectangular manner). By doing so, no adjustments need to be made to an image-based glyph to apply the text. Instead, changes can be made to the glyph, and the edges automatically calculated for wrapping text. The glyph can be changed entirely, and image-based adjustments can still be avoided. This allows a user to continually edit a document, including the object glyph on which text is wrapped instantaneously, and without the need for an additional image-editing application. Applying the wrap around live text itself can yield a more refined, professional look, without the extensive need for image editing required by many of the conventional methods.

(25) One such method that solves many of these problems wraps text around a text-based object, e.g., a glyph generated by glyph data from a font file. That is, the text is wrapped around itself and provides the user with a continuous text wrapping by calculating the edges of the glyph around which the text is wrapped.

(26) To do so, a glyph is generated. The glyph is generated in a document by a document editor, such as a word processing software. The glyph is generated by accessing glyph data within a font file. The glyph data is used to calculate the glyph when rendered on the document by the document editor. The document includes text tiles that are locations of the document where the document editor can insert and maintain text. The text tiles identity text locations within the document where text is located or can be located.

(27) An intersection between the glyph and the text tile is calculated using the glyph data of the

font file. The edges of the glyph are calculated from the glyph data. A location where the edge of the glyph intersects a text tile is identified as the intersection.

(28) Text can then be inserted within the text tile based on the intersection. That is, an insertion location within the text tile can be determined based on the intersection. For instance, the text insertion location may be spaced apart from the intersection location within the text tile. The text is inserted at the insertion location.

(29) Advantageously, as noted, the methods described herein allow a user to wrap text around a text-based glyph in such a manner that the text-based glyph can be easily modified, and the wrapping automatically applied to the modified glyph without the need for significant image processing techniques.

(30) It will be realized that the method previously described is only an example that can be practiced from the description that follows, and it is provided to more easily understand the technology and recognize its benefits. Additional examples are now described with reference to the figures.

(31) With reference now to FIG. **1**, an example operating environment **100** in which aspects of the technology may be employed is provided. Among other components or engines not shown, operating environment **100** comprises server **102**, computing device **104**, and database **106**, which are communicating with document editor **110** via network **108**.

(32) Database **106** generally stores information, including data, computer instructions (e.g., software program instructions, routines, or services), or models used in embodiments of the described technologies. Although depicted as a single database component, database **106** may be embodied as one or more databases or may be in the cloud. In aspects, database **106** is representative of a distributed ledger network. Some computer storage devices suitable for use by database **106** to store information are further described with respect to FIG. **16**.

(33) Network **108** may include one or more networks (e.g., public network or virtual private network [VPN]), as shown with network **108**. Network **108** may include, without limitation, one or more local area networks (LANs) wide area networks (WANs), or any other communication network or method.

(34) Generally, server **102** is a computing device that implements functional aspects of operating environment **100**, such as one or more functions of document editor **110** to wrap text using glyph data. One suitable example of a computing device that can be employed as server **102** is described as computing device **1600** with respect to FIG. **16**. In implementations, server **102** represents a back-end or server-side device.

(35) Computing device **104** is generally a computing device that may be used to edit documents using document editor **110**. As with other components of FIG. **1**, computing device **104** is intended to represent one or more computing devices. One suitable example of a computing device that can be employed as computing device **104** is described as computing device **1600** with respect to FIG. **16**. In implementations, computing device **104** is a client-side or front-end device. In addition to server **102**, computing device **104** may implement functional aspects of operating environment **100**, such as one or more functions of document editor **110**. It will be understood that some implementations of the technology will comprise either a client-side or front-end computing device, a back-end or server-side computing device, or both executing any combination of functions from document editor **110**, among other functions.

(36) As noted, computing device **104** may employ aspects of document editor **110** to wrap text around a text-based glyph. In doing so, text may be wrapped closely to the glyph based on determining where the glyph intersects text tiles where text can be placed, as will be further described.

(37) By way of example only, and with reference to FIGS. **2**A and **2**B, FIG. **2**A and FIG. **2**B provide contrasting illustrations that show examples of text wrapping. FIG. **2**A presents wrapped text around glyph **202**, illustrated as an “A.” This illustrates one example of an image-based text

wrapping where glyph **202** is an image that is defined by theoretical boundary **204**. Text **206** is wrapped around theoretical boundary **204**.

(38) In contrast, FIG. **2**B illustrates glyph **208** with text **212** wrapped around glyph edge **210** based on calculating glyph edge **210** and determining where it intersects text tiles within which text **212** is located.

(39) Computing device **104** may launch document editor **110** and start or retrieve document **120** in which glyphs, such as those illustrated in FIG. **2**B, may be rendered. In general, document editor **110** allows computing device **104** to generate, e.g, apply or edit, text in a document, such as document **120** stored in database **106**. Document editor **110** may include word processors or graphic design applications, and provide tools for manipulating and formatting glyphs. Using document editor **110**, computing device **104** can change the font, select specific glyphs, adjust their size or color, and apply other formatting options to enhance the visual representation of text within a document.

(40) In general, a glyph is a text-based character that can be generated and positioned within document **120**. Glyph generator **112** is employed to generate a glyph. A glyph refers to the visual representation of a character or symbol that is text based. Text is generally used to mean one or more glyphs. When rendered, a glyph is the specific shape or image that appears on document **120**. A glyph can be generated in response to an input, such as rendering a character corresponding to a character on a keyboard used as an input. Other inputs may be used to prompt generation of glyphs as well.

(41) To generate a glyph, glyph generator **112** accesses font file **122** to retrieve glyph data. Font file **122** is a file that comprises glyph data corresponding to a particular set of glyphs. Font file **122** may be in TrueType (.ttf) or OpenType (.otf) format, which contains glyph data about the particular font, including the outlines or shapes of each glyph, although other file types are contemplated and may be usable. As an example, an OpenType font file contains glyph data, in table format, used for rendering text-based glyphs. Portions of the data are used by applications to calculate the layout of text using the font, e.g., the selection of glyphs and their placement within a text tile of a document **120**, which will be further described.

(42) Glyphs, as used herein, are text based. As an example, text-based glyphs can be generated as a Unicode character set, such as UTF-8. When generating the glyph for rendering in a document, such as document **120**, inputs are converted into Unicode values. Each glyph corresponds to a specific Unicode code point, which represents a particular character or symbol. Each glyph corresponds to a specific character, such as a letter, number, punctuation mark, or special symbol, such as an emoji or other similar text-based representation. Glyphs can have different sizes, styles, and appearances within a font, as determined from font file **122**. For example, a font may have multiple versions of the same character with variations in weight (e.g., bold, regular, italic) or width (e.g., condensed, expanded). These different glyph variations allow for the rendering of text in various styles and typographic effects.

(43) The Unicode values are mapped to glyph indices within font file **122**. This mapping is done through tables and encoding schemes defined in font file **122**. Font file **122** provides the information to associate each Unicode character with its corresponding glyph index. Glyph generator **112** generates the visual representation of the glyph by drawing on document **120**. This process may involve rasterizing or vectorizing the glyph outlines, applying font-specific hinting instructions (if present), and rendering them within document **120** having attributes such as color, size, and style. Glyph data within font file **122** may include control points, curves, and other instructions that define the outline of the glyph, such as Bézier Curves, which are mathematical curves defined by the control points and handles, and may be used by glyph generator **112** to generate the visual representation of the glyph within document **120**.

(44) Turning briefly to document **120**, in a specific example, document **120** is a Word document, e.g., a .doc or .docx file. However, it is contemplated that document **120** could include other

document types in which glyphs are rendered. Some document types that may also be suitable include those from plain text (.txt), rich text format (.rtf), tabular documents (e.g., .xls, .xlsx) or CSV (comma-separated values), presentations (e.g., .ppt, .pptx, .odp), and the like.

(45) Document **120** comprises text tiles, sometimes referred to as "text area boundaries." In the context of word processing, a text tile, e.g., a text area boundary, typically refers to the defined limits within which text, including individual glyphs, is located or can be located. It represents the area where text can be entered, edited, or displayed within a document. The text tiles determine the visible space and layout of the text within document **120**.

(46) The text tile may be determined by various factors, including the margins, page size, and other formatting options provided by document editor **110**. The boundaries of the text tiles are established layout settings within document editor **110**, which may define the available width and height for text placement. In some cases, text tiles are background locations and not visible as part of document **120** when rendered on a display device of computing device **104**.

(47) Generally, text tiles maintain organization and readability of the document containing the text within the defined area. They may also aid in determining where line breaks, page breaks, and other organizational features occur. The boundaries of text tiles can be adjusted by modifying the page setup options, adjusting margins, or inserting additional elements like text boxes or image-based objects.

(48) FIGS. **3**A-**3**C provide various illustrations of document **300** showing text tiles with corresponding text located therein. FIG. **3**A illustrates document **300** having text tiles. As illustrated, the text tiles are boundaries that are, in this example, defined by the margins of document **300**. The text tiles illustrate locations within which text can be located. Text tile **302** is one specific example within the text tiles. In general, document **300** may have any number of text tiles, which may be in any orientation based on configurable features within document editor **110**, as previously noted. This is just one example to aid in illustrating example text tiles.

(49) FIG. **3**B illustrates the same document **300**, but includes text located within the text tiles. In this example, and in many cases, each text tile defines the boundaries for a single line of text. A text tile may extend in a direction across which the text of a particular language flows. For instance, in the English language, a text tile may extend horizontally to define the location across which text is placed when inserted. Said differently, the upper and lower boundaries of the text tile may be defined by the text font size, while the lateral boundaries may be defined by margins or other objects within document **300**. In this case, the text may extend within the text tile in a single line between the lateral boundaries. As shown, text tile **302** includes text **304**. As noted, the text tiles define the overall structural layout of the document with respect to where text is located or can be located.

(50) When viewing the document rendered at an interface, the text tiles may not be visible. That is, the boundaries of the text tiles illustrated and described in the figures could be theoretical in nature. A text tile may be defined based on its location, but not visible when rendering the document. That is, a text tile may also include an algorithm that instructs document editor **110** as to the location within a document where text is to be located. Turning to FIG. **3**C, the text, such as **304**, is illustrated without a visualization of the text tiles boundaries. The text assumes the structure defined by the text tiles when located within the document.

(51) To wrap text around a text-based glyph, location determiner **114** can be employed to determine a location used for inserting text relative to a generated glyph. For instance, location determiner **114** can be used to determine, from the glyph data of font file **122**, an intersection between a glyph and a text tile that is used to position text near the glyph in a manner that appears to wrap the text around the text-based glyph. Location determiner **114** can further be employed to determine an insertion location within the text tile based on the intersection, which will be used by other components of document editor **110** to insert the text that is wrapped around the glyph.

(52) To determine the location of the intersection between a glyph and a text tile, location

determiner **114** (e.g., in coordination with glyph generator **112**) can determine the location of the edges, e.g., the outline, of a generated glyph from the glyph data. As noted, the glyph data may include control points, curve segments, and Bézier curve instructions. These instructions provide document editor **110** the necessary information to render the glyph. By processing the control points, curve segments, and Bézier curve instructions according to the configurable document editor features, such as font size, applied font characteristic (bold, italics, underlining, etc.), color, and the like, the edge locations of the glyph are known. As an example, to calculate the glyph edge, a font file (e.g., a .ttf or .otf file), such as font file **122**, may be loaded. The glyph is parsed via the table provide in the font file to extract the shape of the glyph. Quadratic curves are determined using the shape of the glyph and the coordinates of the glyph, thus identifying the glyph edges.

(53) Location determiner **114** can further be employed to determine the location, e.g., the boundaries, of a text tile. In general, text tiles may include boundaries that can be defined based on settings in document editor **110**. For instance, text tiles may have boundaries defined by the margins, or other objects on the document. This provides a relative location of each text tile in the document. For instance, a left and right margin set at one inch may respectively define the left and right horizontal boundary locations of the text tile. The vertical boundaries, upper and lower, may be defined based on font size, paragraph spacing, and other vertical features in the document. Thus, the vertical boundary locations for a text tile may be determined based on the relative position of the vertical features in the document. For instance, the location of an upper boundary of a text tile may be positioned below another text tile having a known vertical distance that is immediately below a one-inch header. Based on the aggregate vertical length of these features, the location of the upper boundary is identified. The lower boundary location can be determined by the font size for text located, or to be located, within the text tile. In some cases, this may be defined using an x-y coordinate positioning system on the document. In one example, a text tile may be calculated by adding the bounding boxes of each glyph. The bounding box may be the maximum area used by a glyph. The text tile is represented as a rectangle in 2D space. The intersection of the 2D text tile can then be determined.

(54) Having determined the location of the glyph and the location of a text tile, an intersection of the glyph and the text tile can be determined by location determiner **114**. For instance, where the location of a text tile boundary and a location of a glyph edge are the same, the location is identified as the intersection between the glyph and the text tile. This may be done for any number of text tiles within a document.

(55) To illustrate, FIGS. **4**A-**4**B are provided to show document portion **400** having a glyph **402** and text tile **404** for which intersection **406** has been determined using location determiner **114**. As illustrated, text tile **404** comprises upper boundary **408** and lower boundary **410**. In some cases, a first boundary of a text tile may be defined based on a second boundary of the text tile intersecting a glyph. In this example, lower boundary **410** has intersected glyph **402** at a location different from upper boundary **408**. As such, upper boundary **408** has been defined based on the intersection of intersection **406** with glyph **402**. In some cases, an intersection may define a new lateral boundary for a text tile, where the new lateral boundary corresponds in location with the intersection. In the example illustrated, lateral boundary **412** is defined based on intersection **406**. The new lateral boundaries determined based on the intersection location are used to define the text tile where text can be located. As such, when text is located within the text tile, the text has the appearance of closely wrapping around a glyph, such as glyph **402**.

(56) Location determiner **114** may further identify an insertion location, which identifies a location where text can be inserted within a text tile. The insertion location can be determined based on the intersection of the text tile and the glyph. As an example, location determiner **114** determines an insertion location based on a lateral boundary of the text tile. The insertion location may be identified by location determiner **114** by spacing the location insertion a distance away from the intersection between the text tile and the glyph at which the lateral boundary is located. Depending

on the word processing program, this spacing may be a defined number of pixels or a defined length from the intersection. In an aspect, the distance at which the insertion location is spaced from the intersection is a configurable option, allowing the user to adjust the distance, thereby providing a mechanism by which a user, via computing device **104**, can adjust how tightly wrapped the text appears around the glyph.

(57) FIG. **4**B illustrates one example. FIG. **4**B illustrates glyph **402** of FIG. **4**A relative to text tile **404**. Here, location determiner **114** has placed insertion location **414** within text tile **404** and spaced apart from intersection **406**. Text **416** can be placed at insertion location **414** and extend within text **416** in a direction away from intersection **406** or lateral boundary **412**, as will be further discussed with reference to text inserter **118**.

(58) As noted, documents may have any number of text tiles. This allows text to be placed at various locations within the document. For instance, text can be placed within vertically stacked text tiles to create horizontal lines of text across a document, as text is placed within each text tile extending from lateral boundary to lateral boundary. In such cases, text can be placed at different insertion locations for different text tiles that are each determined based on their respective intersections with a glyph. This gives the appearance of wrapping around the glyph as text is inserted within the document.

(59) An example of this is illustrated in FIGS. **5**A-**5**B. FIG. **5**A shows document portion **500**, which includes glyph **502**. Here, first text tile **504** intersects glyph **502** at first intersection **506**. First intersection **506** may be determined using methods previously described. That is, first intersection **506** may be determined from the glyph data within a font file from which glyph **502** was generated. As previously described, first lateral boundary **508** of first text tile **504** may be determined based on first intersection **506**.

(60) In this example, a second text tile **510** is placed immediately below first text tile **504** in the document when text tiles are vertically stacked within the document, as illustrated in document portion **500**. For example, second text tile **510** may be positioned vertically adjacent to and parallel with first text tile **504**. Here, second text tile **510** intersects first intersection **506** at second intersection **512**, identifying the location of second lateral boundary **514**, which can be determined using methods previously described. Due to the structure of the edges of glyph **502**, first lateral boundary **508** and second lateral boundary **514** are horizontally offset. Horizontally offset generally refers to a document, such as one from which document portion **500** is derived, that extends in the horizontal direction from a first lateral side to a second lateral side. Objects, such as first lateral boundary **508** and second lateral boundary **514**, are horizontally offset when they measure different distance from either lateral side of the document. Thus, for instance, first lateral boundary **508** may be relatively closer to a first lateral side of a document compared to second intersection **512**, and thus, the two are said to be horizontally offset from one another.

(61) As shown in FIG. **5**B, location determiner **114** can determine the locations of any insertion locations within text tiles. As illustrated, first insertion location **516** has been identified within first text tile **504**, while second insertion location **518** has been identified within second text tile **510**. This can be done using methods previously described, such as determining the insertion locations based on a distance from first intersection **506** or first lateral boundary **508**, and from second intersection **512** or second lateral boundary **514** respectively. Based on the structure of **502**, as determined from its corresponding glyph data, first insertion location **516** and second insertion location **518** are horizontally offset.

(62) Text can be inserted at the insertion locations. Here, text **520** is inserted at first insertion location **516** and extends away therefrom within first text tile **504**. Additionally, additional text **522** is inserted at second insertion location **518** and extends away therefrom within second text tile **510**. Additional text tiles beyond those illustrated in FIG. **5**A and FIG. **5**B may be vertically stacked and text inserted within, thus wrapping the text around glyph **502**.

(63) Some glyph geometries, e.g., the structure defined by an edge of the glyph, create various

geometrical regions that can allow text to be wrapped in various manners. Region classifier **116** may be employed to classify the regions of a glyph so that text can be inserted according to the classified region.

(64) In some cases, region classifier **116** may classify geometrical regions of a glyph based on the glyph data. That is, the glyph data may include information identifying certain geometrical features, such as closed interiors, hooks, and the like. In such cases, region classifier **116** may classify a region of a glyph in a manner corresponding to that identified by the glyph data. Some example regions include inner regions and outer regions. An inner region comprises a location within a glyph that is at least partially enclosed by the glyph. In a specific case, the inner region is entirely enclosed by edges of the glyph. An outer region is an area outside the glyph that is not enclosed by the glyph edges and that extends away from the glyph to an edge of a document that includes the glyph.

(65) In another aspect, regions are identified and classified using the control points and Bézier curve instructions present in the glyph data. That is, these instructions provide glyph generator **112** the information to calculate the positions, shapes, and curves of the edges of the glyph. Knowing the positions, shapes, and curves of the glyph, region classifier **116** can classify the various geometrical regions created by the edges, such as whether an edge encloses or at least partially encloses an area of a document, e.g., an inner region, or whether the edge is an outermost edge of the glyph, identifying the outer region of the glyph, or any other geometrical region that may be classified.

(66) In another aspect, geometrical regions of the glyph may be determined by the intersections of the glyph and text tiles. For instance, if a text tile has a single intersection, then the area in which the textile is located is likely an outer region, since the glyph intersects the text tile at the single location, and the text tile extends away from the glyph toward an edge of a document. If the text tile has more than one intersection with the glyph, the area in which the text tile is located is likely an inner region, since the text tile extends from one edge of the glyph to another edge of the glyph. Put another way, a portion of a text tile intersecting a glyph at a single location may be considered to be within an outer region. A portion of a text tile having two intersections with a glyph may be considered to be within an inner region. Thus, region classifier **116** can use the number of intersections between text tiles and the glyph to determine a geometric region of the glyph, and thus, the text can be inserted into text tiles within the document based on the classified geometrical region.

(67) For instance, using configurable options, text may be inserted within an inner region of a glyph. Text may be inserted into an area that is classified as an outer region of the glyph. In aspects, text may be inserted into any one or more geometrical regions in any combination based on the desired effect selected using document editor **110**.

(68) FIG. **6**A and FIG. **6**B illustrate an example of some geometrical regions for glyph **602**. In FIG. **6**A, the text tiles are shown positioned within outer region **604**. Text tile portion **608** is a portion of a text tile. Text tile portion **608** intersects glyph **602** at a single location, illustrated as intersection **610**. Based on this, region classifier **116** may classify text tile portion **608** as within outer region **604**. Text may be inserted within text tile portion **608** at an insertion location and extend in a direction away from glyph **602**.

(69) FIG. **6**B illustrates text tiles positioned within inner region **606**. Text tile portion **612** is a portion of a text tile. In this example, text tile portion **612** is a portion of the same text tile corresponding to text tile portion **608** in FIG. **6**A. Continuing with FIG. **6**B, text tile portion **612** intersects glyph **602** at two locations, illustrated as first intersection **614** and second intersection **616**. Based on this, region classifier **116** may classify text tile portion **612** as within inner region **606**. In this example, text can be inserted within text tile portion **612** between first intersection **614** and second intersection **616**. Said another way, the inserted text within text tile portion **612** may extend from a first insertion location determined based on first intersection **614** to a second

insertion location determined based on second intersection **616**.

(70) Text inserter **118** may insert text within a document. As noted, text may be inserted at an insertion location and extend within a text tile away from the insertion location at which it is inserted. In an aspect, text is inserted such that the insertion location defines the location of a first character in a line of text within a text tile or portion thereof. In another aspect, text is inserted such that the insertion location defines the location of a last character in a line of text within a text tile or a portion thereof. In this way, text can be wrapped around any side of a glyph or within an inner region of a glyph.

(71) In aspects, text inserter **118** may insert text by generating text from a font file and rendering the text within a text tile. This may be done in a manner similar to that described with glyph generator **112**. In an aspect, a glyph may be generated from a first font file. The inserted text wrapped around the glyph within the text tiles may be generated from a second font file defining a different font style. This allows various stylistic features to be rendered. FIG. **7** illustrates an example using document **700**. Here, glyph **702** and glyph **704** have each been generated from a different font file, and thus, each has a unique font style. The inserted text **706** wrapped around these two glyphs is from a third font file that defines a font style different from those corresponding to glyph **702** and glyph **704**.

(72) In another aspect, the font file may be the same for the glyph and the wrapped text. For example, FIG. **8** illustrates another document **800** comprising glyph **802**. Text **804** has been wrapped around glyph **802**. In this example, glyph **802** and text **804** are the same font style. Here, glyph **802** has been generated from a same font file as text **804**.

(73) In some cases, the glyph for which text has been wrapped around comprises a font size that is relatively larger than that of the wrapped text. Still looking to FIG. **8** as an example, glyph **802** comprises a font size that is relatively larger than that of text **804**.

(74) In an aspect, the font size of a glyph is increased based on the glyph being a first glyph in a text body. A text body may include the entire contents of a document. In an aspect, the text body is a portion of text within a body. This portion may be separated from other text based on features applied within document editor **110**, such as the text body being a portion of text identified as a paragraph, e.g., based on an inserted return; a portion of text identified using a page break; a portion of text identified using a section break; and so forth. Document editor **110** may identify the first glyph within a text body and increase the font size of the glyph relative to text that is wrapped around the glyph. FIG. **8** illustrates an example where document **800** includes text **804**, where text body **806** is a portion of the text determined based on it being a paragraph within document **800**. In some cases, a first glyph is identified, and its corresponding font size is automatically increased relative to the wrapped text. This could be performed in response to a configurable setting applied to document editor **110** using computing device **104**.

(75) FIGS. **9-14** show some examples that can be produced using the technology described herein. FIG. **9** illustrates document **900**. Document **900** comprises glyph **902** around which text **904** has been wrapped. The illustrated text **904** is text that is wrapped within an outer region of glyph **902**. In this case, it extends away from glyph **902**, being inserted using methods previously described, to margin **906** (illustrated using a dashed line) at a lateral end of document **900**.

(76) Another example is illustrated in FIG. **10**. Here, document **1000** comprises glyph **1002**. Text **1004** has been wrapped around glyph **1002**. In this example, text **1004** is wrapped within an inner region of glyph **1002**. As noted, document editor **110** may include configurable settings that, based on an input from computing device **104**, wrap text within an inner region, as illustrated, within an outer region, or both.

(77) FIG. **11** provides yet another example. Document **1100** comprises glyph **1102**, within which text **1104** and text **1106** have been wrapped. In this example, text **1104** and text **1106** are each wrapped within an inner region of glyph **1102**.

(78) FIG. **12** provides another example. Document **1200** comprises glyph **1202** that is wrapped by

text **1204**. In this example, document **1200** also includes glyph **1206**, which is wrapped by text **1208**, and glyph **1210**, which is wrapped by text **1212**. Here, each of glyph **1202**, glyph **1206**, and glyph **1210** are the first glyph in a text body, which are paragraphs in this case. The font size of each of glyph **1202**, glyph **1206**, and glyph **1210** has been increased relative to text **1204**, text **1208**, and text **1212**, respectively. As illustrated, document editor **110** may automatically increase the font size of the first glyph so that the upper boundary of the glyph edges corresponds with the beginning of the text body, while the lower boundary of the glyph edge corresponds to the end of the text body, creating the symmetry illustrated in FIG. **12**.

(79) FIG. **13** is another example. Document **1300** includes glyph **1302**, which comprises an emoji, illustrated as an umbrella in this case. Glyph **1302** has been wrapped generally by text **1304**.

(80) With reference now to FIG. **14**, a block diagram is provided that illustrates an example method **1400** for wrapping text around a glyph. At block **1402** a glyph is generated within a document. This may be performed using glyph generator **112** of document editor **110**. In embodiments, the glyph is generated using glyph data within a font file. The document comprises text tiles, which include locations where text is or can be placed within a document.

(81) At block **1404**, an intersection between a text tile and the glyph is determined. This may be determined using location determiner **114**. For instance, the text tile locations may be known based on document features applied to the document using document editor **110**, such as headers, footers, document margins, paragraph spacing, line spacing, font size, and so on. The location of the glyph, e.g., the location of the glyph edges, can be determined from the glyph data of the font filed. Using this information, the intersections can be determined.

(82) Based on the intersection of the glyph and the text tile, an insertion location may be determined using location determiner **114**. The insertion location may be spaced apart from the intersection location and be located within the text tile. The insertion location may be theoretical in nature and identify a particular location within the document where text can be inserted to wrap around the glyph.

(83) At block **1406**, text is inserted at the insertion location. The text can be inserted within the text tile at the insertion location and extend away from the intersection. This provides the effect of wrapping the text around the glyph. The text can be inserted using text inserter **118**.

(84) In an aspect, geometrical regions of the glyph are classified. Using a computing device, such as computing device **104**, document editor **110** can be employed to provide options for wrapping the text based on the geometrical region. For example, a text tile may lie within an outer region and the text wrapped around the glyph within the outer region. In another aspect, the text tile may lie within an inner region, and the text wrapped around the glyph within the inner region. In aspects of the technology, text is wrapped around the glyph in a combination of regions, such as both an inner and outer region.

(85) In an aspect, the first glyph of a text body, such as a page, paragraph, bullet point, heading section, and the like, may be the glyph around which text is wrapped. The glyph may be generated from the same font file as the wrapped text, or may be generated from a different font file to provide a glyph comprising a different font style relative to the wrapped text.

(86) In some cases, the glyph may comprise a font size that is relatively larger than that of the wrapped text. The font size may be automatically increased by document editor **110** relative to the text. In one example, the font size is increased to match a height of the text body wrapping the glyph. For instance, as vertical lines of text are wrapped, increasing the height of the text body, document editor **110** automatically increases the font of the glyph so that the height of the glyph is about equal to the height of the text body defined by the lines of text, e.g., a plurality of stacked text tiles comprising the wrapped text.

(87) With reference now to FIG. **15**, FIG. **15** illustrates another block diagram having an example method **1500** for wrapping text within a document. In an aspect, method **1500** is performed in addition to method **1400**.

(88) At block **1502**, a second intersection between the glyph and a second text tile is determined. This may be done by location determiner **114**. The second text tile may be vertically stacked with the first text tile. The second text tile may be below or immediately below the first text tile. For instance, the second text tile may be vertically adjacent to the first text tile. The second text tile identifies a location where text is or can be placed. Put another way, a lower boundary of the first text tile may be adjacent to an upper boundary of the second text tile.
(89) At block **1504**, a second insertion location is determined. This may be done using location determiner **114**. The second insertion location may be within the second text tile and spaced apart from the second intersection. The second text tile may be horizontally offset from the first text tile relative to lateral sizes of the document.
(90) At block **1506**, additional text is inserted into the second text tile at the second insertion location. The additional text may be inserted in the second text tile based on text within the first test tile extending from the insertion location to an end of the text tile, e.g., a margin of the document, or to another insertion location within the text tile. This gives the effect of multiple lines of text wrapping around the glyph.
(91) With reference back to FIG. **14** and FIG. **15**, block diagrams are provided and respectively illustrate methods **1400** and **1500** for wrapping text around a glyph. Each block of methods **1400** and **1500** may comprise a computing process performed using any combination of hardware, firmware, or software. For instance, various functions can be carried out by a processor executing instructions stored in memory. The methods can also be embodied as computer-usable instructions stored on computer storage media. The methods can be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few possibilities. Methods **1400** and **1500** may be implemented in whole or in part by components of operating environment **100**.
(92) Having described an overview of some embodiments of the present technology, an example computing environment in which embodiments of the present technology may be implemented is described below in order to provide a general context for various aspects of the present technology. Referring now to FIG. **16** in particular, an example operating environment for implementing embodiments of the present technology is shown and designated generally as computing device **1600**. Computing device **1600** is but one example of a suitable computing environment and is not intended to suggest any limitation as to the scope of use or functionality of the technology. Computing device **1600** should not be interpreted as having any dependency or requirement relating to any one or combination of components illustrated.
(93) The technology may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions, such as program modules, being executed by a computer or other machine, such as a cellular telephone, personal data assistant or other handheld device. Generally, program modules, including routines, programs, objects, components, data structures, etc., refer to code that performs particular tasks or implements particular abstract data types. The technology may be practiced in a variety of system configurations, including handheld devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The technology may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.
(94) With reference to FIG. **16**, computing device **1600** includes bus **1610**, which directly or indirectly couples the following devices: memory **1612**, one or more processors **1614**, one or more presentation components **1616**, input/output (I/O) ports **1618**, input/output components **1620**, and illustrative power supply **1622**. Bus **1610** represents what may be one or more buses (such as an address bus, data bus, or combination thereof). Although the various blocks of FIG. **16** are shown with lines for the sake of clarity, in reality, delineating various components is not so clear, and metaphorically, the lines would more accurately be grey and fuzzy. For example, one may consider

a presentation component, such as a display device, to be an I/O component. Also, processors have memory. The inventors recognize that such is the nature of the art, and reiterate that the diagram of FIG. **16** is merely illustrative of an example computing device that can be used in connection with one or more embodiments of the present technology. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “handheld device,” etc., as all are contemplated within the scope of FIG. **16** and with reference to “computing device.”

(95) Computing device **1600** typically includes a variety of computer-readable media. Computer-readable media can be any available media that can be accessed by computing device **1600** and includes both volatile and non-volatile media, and removable and non-removable media. By way of example, and not limitation, computer-readable media may comprise computer storage media and communication media. Computer storage media, also referred to as a communication component, includes both volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules, or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology; CD-ROM, digital versatile disks (DVDs), or other optical disk storage; magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices; or any other medium that can be used to store the desired information and that can be accessed by computing device **1600**. Computer storage media does not comprise signals per se.

(96) Communication media typically embodies computer-readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media, such as a wired network or direct-wired connection, and wireless media, such as acoustic, radio frequency (RF), infrared, and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

(97) Memory **1612** includes computer-storage media in the form of volatile or non-volatile memory. The memory may be removable, non-removable, or a combination thereof. Example hardware devices include solid-state memory, hard drives, optical-disc drives, etc. Computing device **1600** includes one or more processors that read data from various entities, such as memory **1612** or I/O components **1620**. Presentation component(s) **1616** presents data indications to a user or other device. Example presentation components include a display device, speaker, printing component, vibrating component, etc.

(98) I/O ports **1618** allow computing device **1600** to be logically coupled to other devices, including I/O components **1620**, some of which may be built in. Illustrative components include a microphone, joystick, game pad, satellite dish, scanner, printer, wireless device, etc. The I/O components **1620** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition, both on screen and adjacent to the screen, as well as air gestures, head and eye tracking, or touch recognition associated with a display of computing device **1600**. Computing device **1600** may be equipped with depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB (red-green-blue) camera systems, touchscreen technology, other like systems, or combinations of these, for gesture detection and recognition. Additionally, the computing device **1600** may be equipped with accelerometers or gyroscopes that enable detection of motion. The output of the accelerometers or gyroscopes may be provided to the display of computing device **1600** to render immersive augmented reality or virtual reality.

(99) At a low level, hardware processors execute instructions selected from a machine language

(also referred to as machine code or native) instruction set for a given processor. The processor recognizes the native instructions and performs corresponding low-level functions relating, for example, to logic, control, and memory operations. Low-level software written in machine code can provide more complex functionality to higher levels of software. As used herein, computer-executable instructions includes any software, including low-level software written in machine code; higher level software, such as application software; and any combination thereof. In this regard, components for wrapping text around a glyph can manage resources and provide the described functionality. Any other variations and combinations thereof are contemplated within embodiments of the present technology.

(100) With reference briefly back to FIG. **1**, it is noted and again emphasized that any additional or fewer components, in any arrangement, may be employed to achieve the desired functionality within the scope of the present disclosure. Although the various components of FIG. **1** are shown with lines for the sake of clarity, in reality, delineating various components is not so clear, and metaphorically, the lines may more accurately be grey or fuzzy. Although some components of FIG. **1** are depicted as single components, the depictions are intended as examples in nature and in number and are not to be construed as limiting for all implementations of the present disclosure. The functionality of operating environment **100** can be further described based on the functionality and features of its components. Other arrangements and elements (e.g., machines, interfaces, functions, orders, and groupings of functions, etc.) can be used in addition to or instead of those shown, and some elements may be omitted altogether.

(101) Further, some of the elements described in relation to FIG. **1**, such as those described in relation to document editor **110**, are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein are being performed by one or more entities and may be carried out by hardware, firmware, or software. For instance, various functions may be carried out by a processor executing computer-executable instructions stored in memory, such as database **106**. Moreover, functions of document editor **110**, among other functions, may be performed by server **102**, computing device **104**, or any other component, in any combination.

(102) Referring to the drawings and description in general, having identified various components in the present disclosure, it should be understood that any number of components and arrangements might be employed to achieve the desired functionality within the scope of the present disclosure. For example, the components in the embodiments depicted in the figures are shown with lines for the sake of conceptual clarity. Other arrangements of these and other components may also be implemented. For example, although some components are depicted as single components, many of the elements described herein may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Some elements may be omitted altogether. Moreover, various functions described herein as being performed by one or more entities may be carried out by hardware, firmware, or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. As such, other arrangements and elements (e.g., machines, interfaces, functions, orders, and groupings of functions, etc.) can be used in addition to or instead of those shown.

(103) Embodiments described above may be combined with one or more of the specifically described alternatives. In particular, an embodiment that is claimed may contain a reference, in the alternative, to more than one other embodiment. The embodiment that is claimed may specify a further limitation of the subject matter claimed.

(104) The subject matter of the present technology is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed or disclosed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies.

Moreover, although the terms “step” or “block” might be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly stated.

(105) For purposes of this disclosure, the word “including,” “having,” and other like words and their derivatives have the same broad meaning as the word “comprising,” and the word “accessing” comprises “receiving,” “referencing,” or “retrieving,” or derivatives thereof. Further, the word “communicating” has the same broad meaning as the word “receiving,” or “transmitting,” as facilitated by software or hardware-based buses, receivers, or “transmitters” using communication media described herein.

(106) In addition, words such as “a” and “an,” unless otherwise indicated to the contrary, include the plural as well as the singular. Thus, for example, the constraint of “a feature” is satisfied where one or more features are present. The term “or” includes the conjunctive, the disjunctive, and both (a or b thus includes either a or b, as well as a and b).

(107) For purposes of a detailed discussion above, embodiments of the present technology are described with reference to a distributed computing environment. However, the distributed computing environment depicted herein is merely an example. Components can be configured for performing novel aspects of embodiments, where the term “configured for” or “configured to” can refer to “programmed to” perform particular tasks or implement particular abstract data types using code. Further, while embodiments of the present technology may generally refer to the distributed data object management system and the schematics described herein, it is understood that the techniques described may be extended to other implementation contexts.

(108) From the foregoing, it will be seen that this technology is one well adapted to attain all the ends and objects described above, including other advantages that are obvious or inherent to the structure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is within the scope of the claims. Since many possible embodiments of the described technology may be made without departing from the scope, it is to be understood that all matter described herein or illustrated by the accompanying drawings is to be interpreted as illustrative and not in a limiting sense.

(109) Some example aspects that can be practiced from the forgoing description include the following:

(110) Aspect 1: A system comprising: at least one processor; and one or more computer storage media storing computer readable instructions thereon that when executed by the at least one processor cause the at least one processor to perform operations comprising: accessing glyph data from a font file for generating text in a document; from the glyph data, determining a first intersection between a glyph and a first text tile identifying text location within the document; determining a first insertion location in the first text tile based on the first intersection of the glyph with the first text tile; inserting text into the first text tile at the first insertion location.

(111) Aspect 2: Aspect 1, further comprising: from the glyph data, determining a second intersection between the glyph and a second text tile, the second text tile below the first text tile in the document and identifying another text location within the document; determining a second insertion location in the second text tile based on the second intersection between the glyph and the second text tile, the second insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data; and inserting additional text into the second text tile at the second insertion location.

(112) Aspect 3: Any of Aspects 1-2, further comprising classifying a geometrical region of the glyph based on the first intersection between the glyph and the first text tile, wherein the text is inserted according to the classified geometrical region.

(113) Aspect 4: Aspect 3, wherein the geometrical region is classified as an outer region based on

the first intersection being a single intersection, and the inserted text begins at the first insertion location and extends within the first text tile away from the first intersection.
(114) Aspect 5: Aspect 3, wherein the geometrical region is classified as an inner region based on the first intersection between the glyph and the first text tile and a second intersection between the glyph and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
(115) Aspect 6: Any of Aspects 1-5, wherein the glyph and the inserted text are both generated from the font file.
(116) Aspect 7: Any of Aspects 1-6, further comprising: identifying the glyph as a first glyph within a text body of the document; and based on the glyph being the first glyph within the text body of the document, automatically increasing a font size of the glyph relative to the inserted text.
(117) Aspect 8: A method performed by one or more processors, the method comprising: generating a glyph in a document comprising text tiles identifying text locations within the document, the glyph generated from glyph data within a font file; from the glyph data, determining a first intersection between the glyph and a first text tile within the document; and inserting text into the text tile at a first insertion location determined based on first intersection of the glyph with the first text tile.
(118) Aspect 9: Aspect 8, The method of claim **8**, further comprising: from the glyph data, determining a second intersection between the glyph and a second text tile, the second text tile below the first text tile in the document; and inserting additional text into the second text tile at a second insertion location determined based on the second intersection between the glyph and the second text tile, the second insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data.
(119) Aspect 10: Any of Aspects 8-9, further comprising classifying a geometrical region of the glyph based on the first intersection between the glyph and the first text tile, wherein the text is inserted according to the classified geometrical region.
(120) Aspect 11: Aspect 10, wherein the geometrical region is classified as an outer region based on the first intersection being a single intersection, and the inserted text begins at the first insertion location and extends within the first text tile away from the first intersection.
(121) Aspect 12: Aspect 10, wherein the geometrical region is classified as an inner region based on the first intersection between the glyph and the first text tile and a second intersection between the glyph and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
(122) Aspect 13: Any of Aspects 8-12, wherein the glyph and the inserted text are both generated from the font file.
(123) Aspect 14: Any of Aspects 8-13, further comprising: identifying the glyph as a first glyph within a text body of the document; and based on the glyph being the first glyph within the text body of the document, automatically increasing a font size of the glyph relative to the inserted text.
(124) Aspect 15: One or more computer storage media storing computer readable instructions thereon that, when executed by a processor, cause the processor to perform operations comprising: generating a glyph in a document comprising text tiles identifying text locations within the document, the glyph generated from glyph data within a font file, and the glyph comprising a first font size; from the glyph data, determining a first intersection between the glyph and a first text tile within the document; and inserting text into the text tile at a first insertion location determined based on first intersection of the glyph with the first text tile, the inserted text comprising a second font size that is less than the first font size.
(125) Aspect 16: Aspect 15, further comprising: from the glyph data, determining a second intersection between the glyph and a second text tile, the second text tile below the first text tile in the document; and inserting additional text into the second text tile at a second insertion location determined based on the second intersection between the glyph and the second text tile, the second

insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data, the additional text comprising the second font size.
(126) Aspect 17: Any of Aspects 15-16, further comprising classifying a geometrical region of the glyph based on the first intersection between the glyph and the first text tile, wherein the text is inserted according to the classified geometrical region.
(127) Aspect 18: Aspect 17, wherein the geometrical region is classified as an outer region based on the first intersection being a single intersection, and the inserted text begins at the first insertion location and extends within the first text tile away from the first intersection.
(128) Aspect 19: Aspect 17, wherein the geometrical region is classified as an inner region based on the first intersection between the glyph and the first text tile and a second intersection between the glyph and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
(129) Aspect 20: Any of Aspects 15-19, wherein the glyph and the inserted text are both generated from the font file.

Claims

1. A system comprising: at least one processor; and one or more computer storage media storing computer readable instructions thereon that when executed by the at least one processor cause the at least one processor to perform operations comprising: accessing glyph data from a font file for generating text in a document; calculating from the glyph data a first intersection between a glyph, having been generated from the glyph data of the font file, and a first text tile identifying text location within the document, the first intersection calculated by identifying a location of the first text tile that is the same as a location on a Bézier curve calculated according to the font file for control points of the glyph; determining a first insertion location in the first text tile based on the first intersection of the glyph with the first text tile; and inserting text into the first text tile at the first insertion location.
2. The system of claim 1, further comprising: from the glyph data, calculating a second intersection between the glyph and a second text tile, the second text tile below the first text tile in the document and identifying another text location within the document; determining a second insertion location in the second text tile based on the second intersection between the glyph and the second text tile, the second insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data; and inserting additional text into the second text tile at the second insertion location.
3. The system of claim 1, further comprising classifying a geometrical region of the glyph based on the first intersection between the glyph and the first text tile, wherein the text is inserted according to the classified geometrical region.
4. The system of claim 3, wherein the geometrical region is classified as an outer region based on the first intersection being a single intersection, and the inserted text begins at the first insertion location and extends within the first text tile away from the first intersection.
5. The system of claim 3, wherein the geometrical region is classified as an inner region based on the first intersection between the glyph and the first text tile and a second intersection between the glyph and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
6. The system of claim 1, wherein the glyph and the inserted text are both generated from the font file.
7. The system of claim 1, further comprising: identifying the glyph as a first glyph within a text body of the document; and based on the glyph being the first glyph within the text body of the document, automatically increasing a font size of the glyph relative to the inserted text.

8. A method performed by one or more processors, the method comprising: generating a glyph in a document comprising text tiles identifying text locations within the document, the glyph generated from glyph data within a font file; from the glyph data, calculating a first intersection between a Bézier curve of the glyph and a first text tile within the document, the Bézier curve calculated according to the font file for control points of the glyph; and inserting text into the text tile at a first insertion location determined based on first intersection of the glyph with the first text tile.
9. The method of claim 8, further comprising: from the glyph data, calculating a second intersection between the Bézier curve and a second text tile, the second text tile below the first text tile in the document; and inserting additional text into the second text tile at a second insertion location determined based on the second intersection between the Bézier curve and the second text tile, the second insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data.
10. The method of claim 8, further comprising classifying a geometrical region of the glyph based on the first intersection between the Bézier curve and the first text tile, wherein the text is inserted according to the classified geometrical region.
11. The method of claim 10, wherein the geometrical region is classified as an outer region based on the first intersection being a single intersection, and the inserted text begins at the first insertion location and extends within the first text tile away from the first intersection.
12. The method of claim 10, wherein the geometrical region is classified as an inner region based on the first intersection between the Bézier curve and the first text tile and a second intersection between the Bézier curve and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
13. The method of claim 8, wherein the glyph and the inserted text are both generated from the font file.
14. The method of claim 8, further comprising: identifying the glyph as a first glyph within a text body of the document; and based on the glyph being the first glyph within the text body of the document, automatically increasing a font size of the glyph relative to the inserted text.
15. One or more computer storage media storing computer readable instructions thereon that, when executed by a processor, cause the processor to perform operations comprising: generating a glyph in a document comprising text tiles identifying text locations within the document, the glyph comprising a Bézier curve generated from glyph data within a font file, and the glyph comprising a first font size; from the glyph data, determining a first intersection between the Bézier curve of the glyph and a first text tile within the document for wrapping text of the first text tile around the glyph, the first intersection calculated by identifying a location of the first text tile that is the same as a location on the Bézier curve; and inserting the text into the text tile at a first insertion location determined based on first intersection of the glyph with the first text tile, the inserted text comprising a second font size that is less than the first font size.
16. The media of claim 15, further comprising: from the glyph data, determining a second intersection between the Bézier curve of the glyph and a second text tile, the second text tile below the first text tile in the document, and the second intersection determined for wrapping additional text of the second text tile around the glyph; and inserting the additional text into the second text tile at a second insertion location determined based on the second intersection between the glyph and the second text tile, the second insertion location being horizontally offset from the first insertion location within the document based on the glyph as determined from the glyph data, the additional text comprising the second font size.
17. The media of claim 15, further comprising classifying a geometrical region of the glyph based on the first intersection between the glyph and the first text tile, wherein the text is inserted according to the classified geometrical region.
18. The media of claim 17, wherein the geometrical region is classified as an outer region based on the first intersection being a single intersection, and the inserted text begins at the first insertion

location and extends within the first text tile away from the first intersection.
19. The media of claim 17, wherein the geometrical region is classified as an inner region based on the first intersection between the glyph and the first text tile and a second intersection between the glyph and the first text tile, and wherein the inserted text extends within the first text tile between the first intersection and the second intersection.
20. The media of claim 15, wherein the glyph and the inserted text are both generated from the font file.